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TRANSPORT PHENOMENA I

FALL 2025

EXAM 1

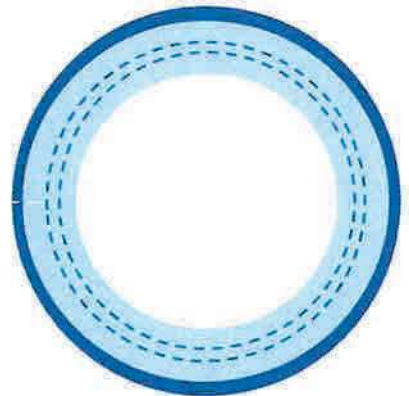
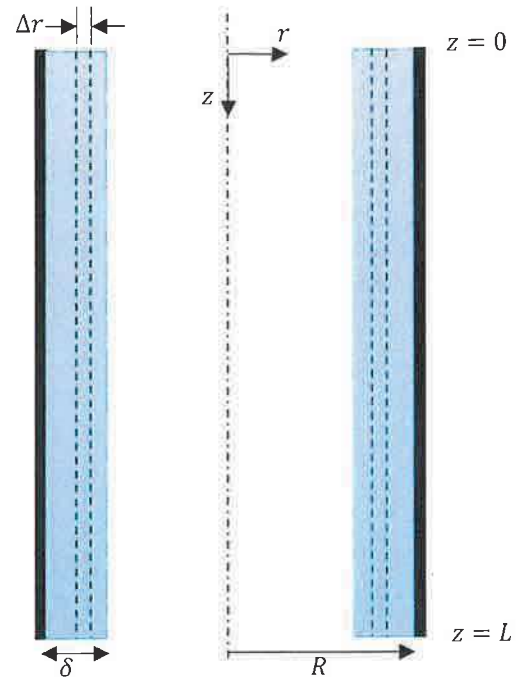
A thin liquid film of thickness δ is flowing down on the inner surface of a long vertical cylindrical tube of inner radius R .

Assumptions:

- Laminar flow
- Newtonian fluid with constant density ρ and constant viscosity μ
- Constant pressure within the liquid: $\frac{\partial p}{\partial z} = 0$
- Steady state flow, where $v_z = v_z(r)$ and $v_r = v_\theta = 0$

Answer the following questions, in terms of parameters δ , R , μ , ρ , and g .

- 1- Write down the steady state z -momentum balance on a cylindrical shell of thickness Δr and length L (represented as the dashed lines in the schematics).
- 2- Apply Newton's law of viscosity to the above momentum balance to derive a second-order differential equation for the velocity profile of the liquid.
- 3- Write down two boundary conditions for this velocity profile.
- 4- Derive the velocity profile by integrating the differential equation of part 2 and evaluating the integration constants.
- 5- Calculate the mass flow rate of the liquid film.
Hint: $\int dx x \log x = \frac{1}{2} x^2 \log x - \frac{1}{4} x^2$ $\dot{\omega} = \rho \dot{Q}$



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$$\frac{\partial p}{\partial z} = 0 \quad v_z = v_z(r), v_r = v_\theta = 0$$

$$|\tau| = -\mu(|\nabla v| + |\nabla v|^T)$$

$$\nabla v_{rz} = \frac{\partial v_z}{\partial r}$$

$$\tau_{rz} = -\mu \left[\frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right]$$

$$\frac{\partial v_z}{\partial r} = \frac{P_0 - P_L}{L}$$

$$P = p - \rho g z$$

$$\frac{\partial}{\partial r} \left(-\mu \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right) = \frac{P_0 - P_L}{L}$$

$$\int_0^L \int_0^{\Delta r}$$

$$\int \frac{\partial}{\partial r} \left(\frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right) = \int \frac{P_0 - P_L}{\mu L} \rho dr$$

$$\int \left(\frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right) = \int \frac{P_0 - P_L}{-\mu L} r dr + C_1$$

$$v_r + v_z = \frac{P_0 - P_L r^2}{-2\mu L} + C_1 r + C_2$$

3. B.C. 1: $r=0$ @ $z=0$

B.C. 2:

$$0 = \frac{P_0 - P_L \cdot 0^2}{-2\mu L} + C_1 r + C_2 \quad C_1 = 0$$

$$\begin{aligned} \{v \cdot \nabla T\}_{rz} &= (v \cdot \nabla) T_{rz} + \frac{v_\theta}{r} (T_{rr} - T_{\theta\theta}) \\ &= \left(v_r \frac{\partial}{\partial r} + \frac{v_\theta}{r} \frac{\partial}{\partial \theta} + v_z \frac{\partial}{\partial z} \right) T_{rz} + \frac{v_\theta}{r} (T_{rr} - T_{\theta\theta}) \\ &= v_z \frac{\partial}{\partial z} T_{rz} + T_{rr} - T_{\theta\theta} \end{aligned}$$

$$T_{rr} = -\mu \left[2 \frac{\partial v_r}{\partial r} \right] + \left(\frac{2}{3} \mu - k \right) \left(\frac{v_\theta}{r} \right)$$

$$T_{\theta\theta} = -\mu \left[2 \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) \right] + \left(\frac{2}{3} \mu - k \right) \left(\frac{v_\theta}{r} \right)$$

$$(\nabla \cdot v) = 0 \quad \rho = \text{const.}$$

$$\begin{aligned} \{v \cdot \nabla T\}_{rz} &= v_z \frac{\partial}{\partial z} - \left(-\mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) \right] + \left(\frac{2}{3} \mu - k \right) \right) \\ &\quad + \left(-\mu \left[2 \frac{\partial v_r}{\partial r} \right] + \left(\frac{2}{3} \mu - k \right) \right) \end{aligned}$$

$$\{v \cdot \nabla T\}_{rz} = v_z \frac{\partial}{\partial z} + \mu \left(2 \frac{1}{r} \right) + \left(\frac{2}{3} \mu - k \right) - 2\mu + \left(\frac{2}{3} \mu - k \right)$$

$$5. \dot{\omega} = \rho \dot{Q}$$

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